# Lab Manual for Digital Logic and Design

## Implementation and Verification of half adder and full adder

# EXPERIMENT NO. : 06

## Implementation and Verification of half adder and full adder

**(a)** Using basic gates

**(b) Using universal gates**

**Part (a)**

### Implementation of half Adder

Objective:

Digital computers perform a variety of information-processing tasks. Among the basic functions encountered are the various arithmetic operations. The most basic arithmetic operation is the addition of two binary digits. In this experiment we'll know about the circuits that will add binary digits.

Apparatus:

- Logic trainer
- Connecting wires
- IC: 7400, 7408, 7432, 7486
- Power supply

Theory:

A combinational circuit that performs the addition of two bits is called a half-adder. This circuit requires two binary inputs and two binary outputs. Let the inputs be A & B.

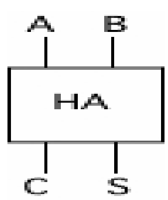


Figure. 6.1: Block Diagram of Half Adder

The circuit diagram for the half adder using basic gates is:

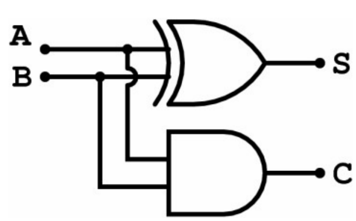


Figure 6.2: Half Adder using basic gates

The circuit diagram for the half adder using universal (Nand) gates is:

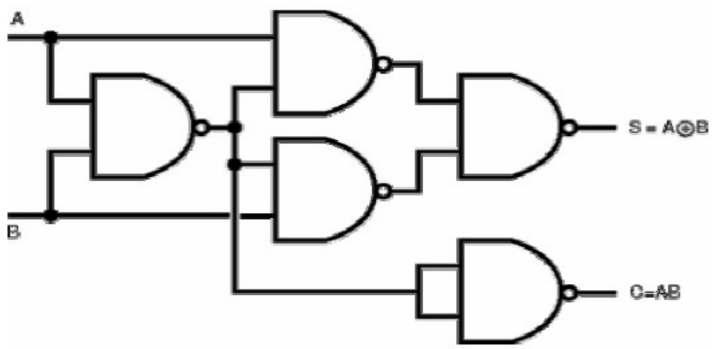


Figure 6.3: Half Adder using universal gates

The low order output will be called "S" because it represents the sum, and the high order output will be called "C" because it represents the carry out. The simplified Boolean functions for the two outputs can be obtained directly from the truth table. The simplified sum of the products expressions are:

$$\mathbf{S = A'. B + A .B'}$$
$$\mathbf{C = A.B}$$

### Procedure:

- Construct the circuit as shown in logic diagram.
- Connect inputs to switches of logic trainer and output to LED.
- Apply various combinations of inputs and observe the conditions of LEDs.
- Mark results in truth table.

### Simulation:

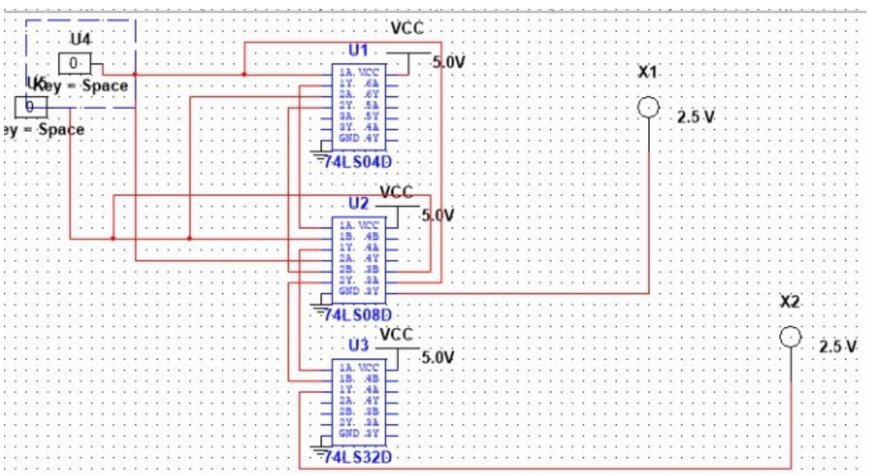


Fig:6.1.1 Simulation of Half adder

### Observations & Calculations:

*Table 6.1: Truth table that verifies half adder*

| A | B | Sum | Carry |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |

| | | | |
|---|---|---|---|
| 1 | 1 | 0 | 1 |

Part (b)

## Implementation of full adder

**Objective:**

Digital computers perform a variety of information-processing tasks. Among the basic functions encountered are the various arithmetic operations. The most basic arithmetic operation is the addition of two binary digits. In this experiment we'll know about the circuits that will add binary digits.

### Apparatus:

- Logic trainer
- Connecting wires
- IC: 7400, 7408, 7432, 7486
- Power supply

### Theory:

A full adder is a combinational circuit that forms the arithmetic sum of three input bits. It consists of three inputs and two outputs. Two of the input variables represent the two significant bits to be added. The third input represents the carry from the previous lower significant position. A full adder can be constructed using two half adders and an OR gate.

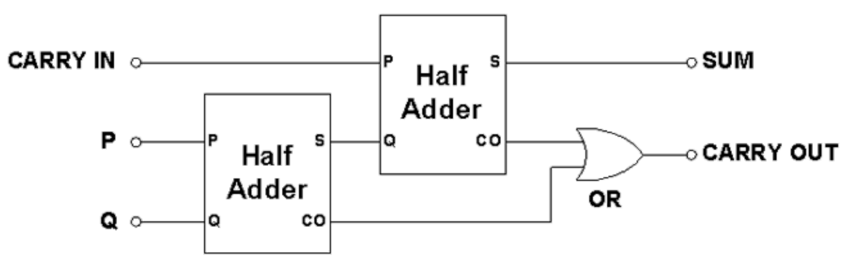


Figure 6.4: Full Adder using two half adders

The circuit diagram for the full adder using basic gates is:

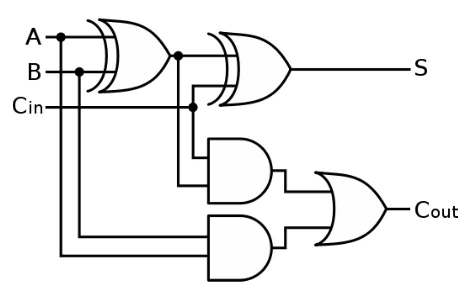


Figure 6.5: Full Adder using basic gates

The circuit diagram for the full adder using universal (Nand) gates is:

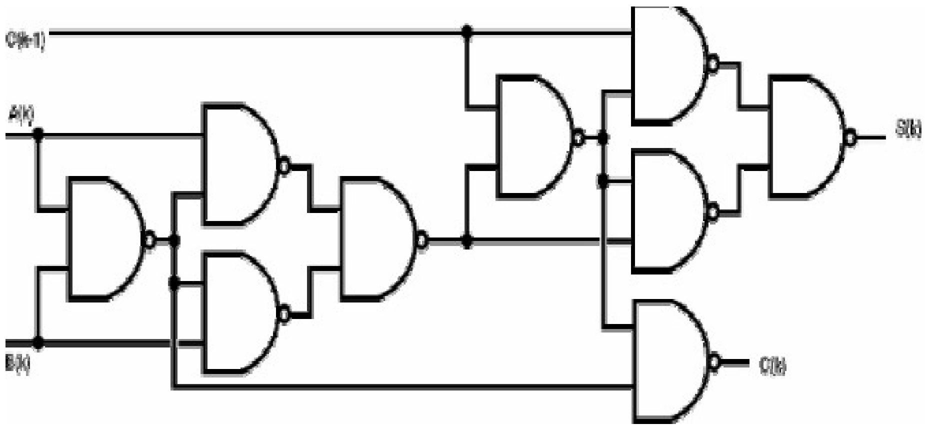


Figure 6.6: Full Adder using universal gates

The two outputs are designated by the symbols S(k) for sum and C(k) for carry. The binary variable S (k) gives the value of the least significant bit of the sum. The binary variable C (k) gives the output carry. The simplified sum of the products expressions are:

**S = A'B'C+A'BC'+AB'C'+ABC**

**C = AB+AC+BC**

Procedure:

- Construct the circuits shown logic diagrams.
- Connect inputs to switches of logic trainer and output to LED.
- Apply various combinations of inputs and observe the conditions of LEDs.
- Mark results in truth table.

Simulation:

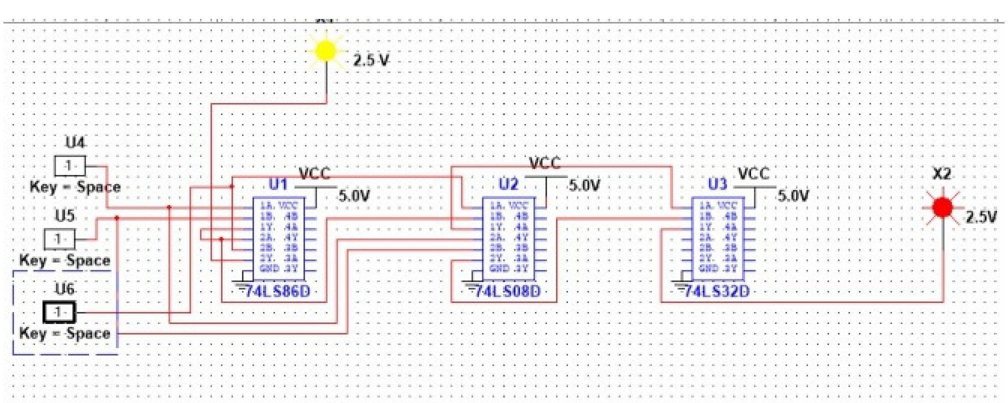


Fig:6.1.1 Simulation of Full Adder

Observations & Calculations:

*Table 6.2: Truth table that verifies full adder*

| A | B | C | Sum (S) | Carry (C) |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

Conclusion:

Implemented the following circuits in MultiSim. The outputs were observed on LED's andoutputs were same as truth table.